

Specification – Industry Style

Class 1: User

Required behaviour

- Each user has a **username** and a **password**.
- You must be able to:
 - Get the username.
 - Check whether a given password is correct.
 - Change the password, but only if the old password is entered correctly.

Required methods

- `public User(String username, String password)`
Creates a new user with the given username and password.
- `public String getUsername()`
Returns the username.
- `public boolean checkPassword(String password)`
Returns true if the given password matches the user's password.
- `public void setPassword(String oldPassword, String newPassword)`
Changes the password only if the old password is correct.
Prints a message to show whether the password was changed successfully.

Fields

- `private final String username`
- `private String password`

Rules

- The username cannot be changed after the user is created.
- Do **not** provide a `getPassword` method.
- Use the `equals` method (not `==`) when comparing strings.
- Keep all fields **private** to ensure encapsulation.

Class 2: BankAccount

Required behaviour

- Each account has:
 - An account number.
 - An account holder (a User object).
 - A current balance.
- You must be able to:
 - Get the account number.
 - Get the account holder.
 - Get the current balance.
 - Deposit money into the account.
 - Withdraw money from the account, up to the current balance.

Required methods

- `public BankAccount(String accountNumber, User accountHolder)`
Creates a new bank account with a zero balance.
- `public String getAccountNumber()`
Returns the account number.
- `public User getAccountHolder()`
Returns the account holder.
- `public double getBalance()`
Returns the current balance.
- `public void deposit(double amount)`
Increases the balance if the amount is positive; otherwise prints “Invalid amount.”
- `public void withdraw(double amount)`
Decreases the balance if the amount is positive and not more than the balance; otherwise prints “Invalid amount.”

Fields

- `private final String accountNumber`
- `private final User accountHolder`
- `private double balance`

Rules

- The account number and account holder cannot be changed after the account is created.
- The balance must never become negative.
- All fields must be **private** to ensure encapsulation.